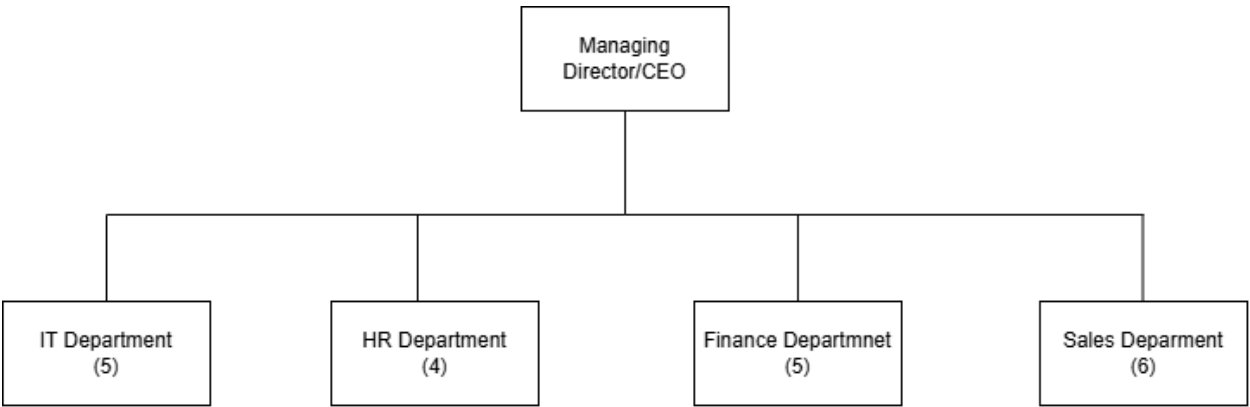


Part 1: Company Profile

- **Company Name:** ABC Startup Solutions
- **Nature of Business:** Software Development & Custom Tech Solutions
- **Vision:** To become a leading provider of innovative, reliable, and scalable software solutions for businesses in Southeast Asia.
- **Mission:** To empower organizations through custom-built software applications while maintaining high standards of system availability, data security, and operational efficiency.
- **Office Location:** 5th Floor, Tech Innovation Tower, Laguna Technopark, Biñan, Laguna, Philippines



Department	Headcount	Key Functions
Information Technology	5	Infrastructure management, software development, system maintenance, tech support
Human Resources	4	Recruitment, employee relations, payroll support, onboarding
Finance	5	Financial planning, accounting, billing, payroll processing
Sales	6	Client acquisition, business development, account management
Total	20	

Part 2: Enterprise Hardware Inventory

Asset ID	Hardware Description	Quantity	Department	Purpose / Justification
HW-DESK-01	High-Performance Workstation (Intel i7, 32GB RAM, 1TB SSD)	5	IT	Heavy software development, virtualization, and testing
HW-LAP-01	Business Laptop (Intel i5, 16GB RAM, 512GB SSD)	15	HR (4), Finance (5), Sales (6)	Mobility for meetings, office productivity, and remote work readiness
HW-MON-01	27" IPS FHD Dual Monitor Setup	10	IT	Dual-screen setup for coding and debugging efficiency
HW-MON-02	24" FHD Single Monitor	15	HR, Finance, Sales	Extended desktop display for administrative office tasks
HW-SRV-01	Rack Server (Dell PowerEdge R450 - Xeon 16-Core, 64GB RAM, 4TB RAID)	1	Server Room (IT)	Local domain controller, file hosting, internal application staging
HW-RTR-01	Enterprise Router (MikroTik Cloud Core Router)	1	Server Room (IT)	Core network routing and inter-VLAN management

Asset ID	Hardware Description	Quantity	Department	Purpose / Justification
HW-SW-01	48-Port Managed Gigabit PoE+ Switch	1	Server Room (IT)	Central network connectivity for all wired devices and access points
HW-AP-01	Wi-Fi 6 Access Point (Ubiquiti Unifi AP)	2	Office Floor	Seamless wireless coverage across all department work areas
HW-PRN-01	Heavy-Duty Network Laser Multifunction Printer	1	Shared Area	High-volume printing, scanning, and copying for all departments
HW-UPS-01	3000VA Smart UPS System	1	Server Room (IT)	Power backup to prevent server and network device downtime during outages
HW-NAS-01	4-Bay Network Attached Storage (16TB Total RAID 5)	1	Server Room (IT)	Centralized local backup storage for company assets and archives
HW-EXT-01	4TB Rugged External Hard Drive	2	IT / Security	Cold offsite physical backup rotation

Part 3: Enterprise and Software Inventory

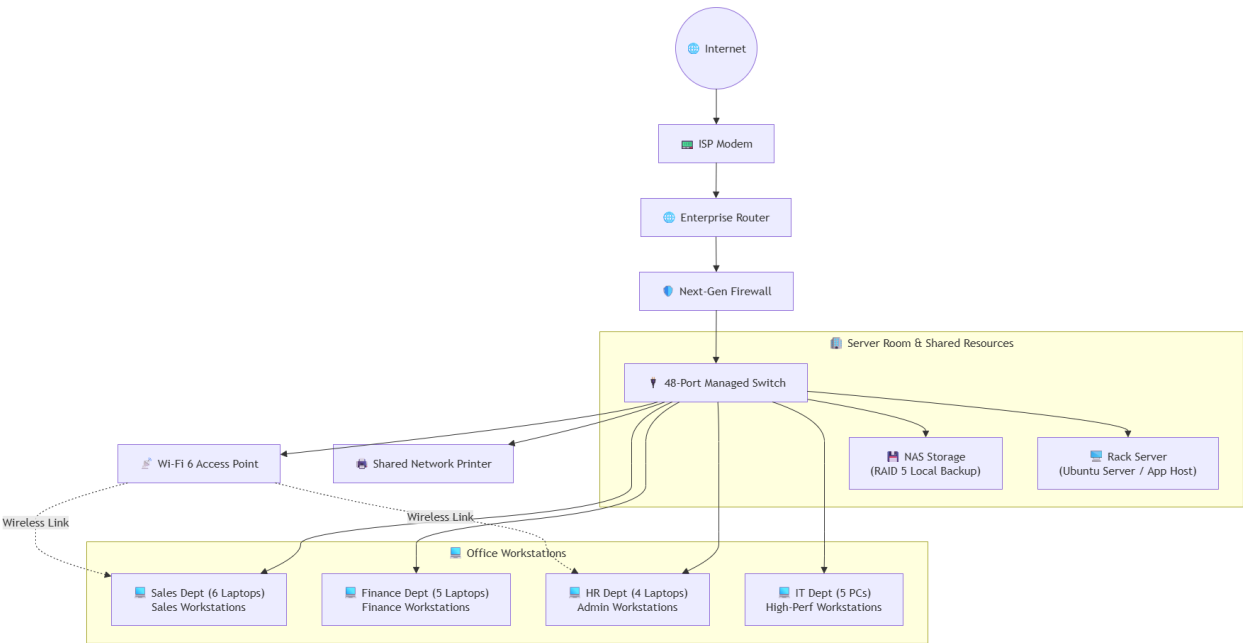
Software	Version	License Type	Purpose / Justification
Windows 11 Pro	22H2 / 23H2	Commercial OEM/Volume	OS for end-user workstations with BitLocker and Domain Join support
Ubuntu Server	22.04 LTS	Open-Source / Free	OS for the central server running local web applications and services
Microsoft 365	Business Standard	Enterprise Subscription	Standard office productivity tools (Word, Excel, Outlook, Teams)
VS Code	Latest Stable	Open-Source / Free	Primary IDE used by the IT development team
Git	Latest Stable	Open-Source / Free	Source code version control for software development workflows
GitHub Desktop	Latest Stable	Open-Source / Free	GUI client to streamline Git operations for developers
VirtualBox	7.x	Open-Source / Free	Local environment isolation and sandbox testing for IT staff
Google Chrome	Latest	Enterprise Freeware	Standard corporate web browser
Microsoft Defender	Integrated Enterprise	Included with OS	Endpoint security, malware protection, and threat response
AnyDesk	Enterprise	Commercial License	Secure remote desktop support tool for internal IT assistance
7-Zip	23.x	Open-Source / Free	Secure file compression and archive handling

Part 4: Enterprise Network Inventory

Equipment Name	Brand / Model	Quantity	Specifications / Details	Purpose
ISP Modem	Fiber Modem (ISP Provided)	1	Dual Band, Fiber Optic Gateway (500 Mbps)	Fiber internet termination point
Router	MikroTik CCR1009	1	8x GbE, 1x SFP+, Hardware Encryption	Gateway routing and Bandwidth management
Firewall	Fortinet FortiGate 60F	1	Next-Gen Firewall (NGFW), SSL VPN	Perimeter network defense, intrusion prevention, deep inspection
Managed Switch	Cisco Catalyst 2960X	1	48-Port GbE PoE+, Layer 2/3 Managed	Central switch managing isolated VLANs per department
Wireless AP	UniFi 6 Long-Range	2	Wi-Fi 6, Dual-Band, PoE Powered	High-speed wireless access for laptops and mobile devices
Patch Panel	48-Port Cat6 Patch Panel	1	1U Rackmount RJ45 Keystone	Structured cabling management in server rack
Cat6 Ethernet Cables	Solid Copper Cat6 (305m Roll)	2 Box	1000Mbps / 10Gbps rated	Structured horizontal cabling through floor/walls

Equipment Name	Brand / Model	Quantity	Specifications / Details	Purpose
RJ45 Connectors	Pass-Through Cat6 Connectors	200 Pcs	Gold-plated contacts	Cable terminations for network drops

Part 5: Enterprise Network Diagram Guidelines



Part 6: System Administration Roles

1. Helpdesk Technician

- Responsibilities: Serves as the first point of contact (Tier 1) for technical support. Manages IT helpdesk tickets, troubleshoots hardware/software issues for end-users, sets up new employee workstations, and maintains device inventory.
- Skills: Operating system troubleshooting (Windows/macOS), basic networking, active listening, customer service, hardware maintenance.
- Common Tools: Zendesk/Jira Service Desk, AnyDesk, Sysinternals, Active Directory Users & Computers.
- Certifications: CompTIA A+, Microsoft Certified: Modern Desktop Administrator Associate.

2. Network Administrator

- Responsibilities: Configures, maintains, and monitors local network infrastructure (switches, routers, firewalls, Wi-Fi APs). Ensures network uptime, manages VLAN segmentation, monitors bandwidth usage, and enforces network security policies.
- Skills: TCP/IP routing, subnetting, firewall configuration, VLAN segmentation, Wi-Fi optimization, network packet analysis.
- Common Tools: Wireshark, Cisco Packet Tracer, PuTTY, SolarWinds Network Performance Monitor.
- Certifications: Cisco Certified Network Associate (CCNA), CompTIA Network+.

3. Linux System Administrator

- Responsibilities: Installs, configures, and maintains Linux-based servers (e.g., Ubuntu Server, RHEL). Manages user permissions, automates routine tasks via shell scripts, monitors server performance, handles system updates, and ensures continuous service uptime.
- Skills: Bash scripting, user/group permission management, SSH security, systemd service administration, web/database server hosting.
- Common Tools: Bash CLI, SSH, Ansible, Cron, htop/top, Webmin.
- Certifications: CompTIA Linux+, Red Hat Certified System Administrator (RHCSA), LPIC-1.

4. Cloud Administrator

- Responsibilities: Provisions, configures, and monitors cloud-based infrastructure (AWS, Azure, or GCP). Manages cloud storage, user access policies (IAM), controls cloud resource deployment costs, and supports hybrid-cloud integrations.
- Skills: Cloud service management, Infrastructure as Code (Terraform), IAM security policies, virtual networking in the cloud, cloud cost management.
- Common Tools: AWS Management Console, Azure Portal, Terraform, AWS CLI, CloudWatch.
- Certifications: AWS Certified Solutions Architect – Associate, Microsoft Certified: Azure Administrator Associate.

Collaboration Summary

In an enterprise environment, these four roles form a unified IT operations ecosystem. The Helpdesk Technician resolves immediate end-user requests and routes network/server-level bugs to Tier 2/3 specialists. The Network Administrator maintains stable connectivity and secure pathways so that the Linux System Administrator can host and optimize local services securely. Meanwhile, the Cloud Administrator extends local capabilities to offsite cloud platforms. Together, they build a cohesive network environment ensuring continuous system availability and end-user productivity.

Part 7: Infrastructure Recommendations

- **Internet Provider (ISP):** Enterprise Fiber Internet Plan (500 Mbps symmetrical upload/download) with a static IP address to guarantee high application availability and stable remote access.
- **Server Specifications:** Dell PowerEdge Rack Server with dual power supplies, 64GB DDR4 RAM, and 4TB RAID 5 storage array to safeguard against hardware drive failures.
- **Backup Strategy:** Implement a strict 3-2-1 Backup Strategy:
 - 3 copies of critical company data.
 - 2 different storage media types (Local Server RAID + Local Synology NAS).
 - 1 offsite/cloud backup location (Encrypted Cloud Storage + Offsite External Drives).
- **Security & Antivirus:** Centralized deployment of Microsoft Defender Enterprise across all endpoints with real-time scanning, network isolation features, and automated threat updates.
- **Password Policy:** Enforce enterprise domain policies requiring a minimum 12-character passphrase length (combining uppercase, lowercase, numbers, and symbols), mandatory Multi-Factor Authentication (MFA), and a quarterly password rotation requirement.
- **Expansion Plan:** The network is architected with scalability in mind. The 48-port switch currently utilizes only 25 ports, allowing seamless growth for up to 23 additional wired devices without requiring hardware upgrades.

Part 8: Personal Reflection

Completing this project taught me a lot about what it actually takes to set up an entire IT infrastructure for a startup. Before this, I thought starting a company's tech setup was just about buying a bunch of computers and connecting them to Wi-Fi. Building the hardware, software, and network inventories showed me how much thought goes into picking the right specs, licenses, and tools for each department so nobody runs out of storage or processing power.

The most challenging task for me was designing the network topology diagram in Draw.io and setting up the inventories. Figuring out how devices connect logically from the internet connection and firewall down to the switches, servers, and department workstations took a lot of trial and error. Planning before deployment is super important because buying equipment without a plan leads to wasted money, security risks, and broken connections. If you don't calculate how many switch ports or IP addresses you need beforehand, you might end up with missing cables, overloaded routers, or unsecure devices. Proper planning makes sure the network can handle growth and stays safe from day one.

This project will help me become a better System Administrator because it gave me hands-on experience thinking like a real IT professional. I learned how to analyze business needs, document every piece of hardware and software, and explain technical choices clearly. It showed me that being a good SysAdmin isn't just about fixing broken computers, but also about building a solid, organized foundation that keeps the whole company running smoothly.